

Chhaya (छाया)



IoT Device Identification via Traffic Fingerprinting

1. Purpose: To identify every device on a network purely from how it communicates — not what it says — and to catch impostors even when all traffic is fully encrypted. Chhaya (Hindi for “shadow”): every device leaves a shadow of itself in its traffic pattern, and that shadow alone identifies it — packet payloads are never read.

2. How It Works:

Passive capture reads only packet metadata (timestamp, size, source IP); a rolling 12-second window per device yields four behavioural features: packet size, inter-packet interval, packet rate, and burstiness.

A Random Forest classifier labels each device (SensorNode, CameraStream, SmartSwitch) with a confidence score every 2.5 seconds on a live dashboard.

3. Detection Layer (Rogue & Spoofing):

Unknown traffic matching no known device profile is flagged with an amber rogue alert within seconds.

An impostor whose behaviour drifts from its own baseline ($z > 4$) triggers a red spoofing alert; attack traffic never reshapes the baseline it is judged against.

4. Live Demonstration (Real Hardware):

A real ESP8266 microcontroller streams UDP packets over Wi-Fi; Chhaya identifies it as SensorNode (confidence 1.00) from radio traffic alone, with a live packet-rate graph.

Re-flashing the same board to impersonate the sensor fools the classifier, but the behaviour detector fires a spoofing alert (z up to 11.8); an unknown phone is flagged rogue in seconds.

5. Real-World Deployment (The Differentiator):

Chhaya is software: it runs on a Raspberry Pi behind the home router, or on a SPAN/mirror port in enterprises — no agents, no per-device setup, works retroactively.

It inspects no content, so encryption is irrelevant: HTTPS stays unreadable while packet sizes and timing still leak the shadow.

6. The total cost of the prototype is approx. Rs. 400 - 500/- (one ESP8266 NodeMCU; all software — Python, scikit-learn, Flask — is open-source) (may vary based on market MRP and may reduce in mass production).